

西南财经大学本科期末考试试题册（B）

（2020—2021学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVA SE）(英)

命题教师：王磊

适用对象（年级专业）：2020 信息管理与信息系统专业

使用试题的任课教师姓名：王磊

试题说明：

- 1、考试类型：闭卷[] 开卷[✓]
- 2、本套试题共 2 道大题，共 7 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 字典[] 等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
1. 出示学生证（或身份证）和准考证于桌面左上角，以备查验。
 2. 拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 3. 答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
 4. 所有答案均需填写在答题纸上，答在试题册上无效。
 5. 考生不得携带任何通讯工具进入考场。
 6. 考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
 7. 严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	2	3	4	5	6	7	8	9	10

1. The expression **(int)(76.0252175 * 100) / 100** evaluates to _____.

- A. 76
- B. 76.0252175
- C. 76.03
- D. 76.02

2. What is y after the following switch statement? _____.

```
int x = 0;
int y = 0;
switch (x + 1) {
    case 0: y = 0;
            break;
    case 1: y = 1;
    default: y = -1
}
```

- A. 2
- B. 1
- C. 0
- D. -1

3. Analyze the following code, which description is **correct** _____.

```
public class Test {
    public static void main (String[] args) {
```

```

int i = 0;
for (i = 0; i < 10; i++);
    System.out.println(i + 4);
}

```

- A. The program compiles despite the semicolon (;) on the for loop line, and displays 4.
- B. The program compiles despite the semicolon (;) on the for loop line, and displays 14.
- C. The program has a compile error because of the semicolon (;) on the for loop line.
- D. The program has a runtime error because of the semicolon (;) on the for loop line.

4. Analyze the following code, which description is correct _____.

```

class Test {

    public static void main(String[] args) {
        System.out.println(xMethod((double)5));
    }

    public static int xMethod(int n) {
        System.out.println("int");
        return n;
    }

    public static long xMethod(long n) {
        System.out.println("long");
        return n;
    }

}

```

- A. The program displays int followed by 5.
- B. The program displays long followed by 5.
- C. The program runs fine but displays things other than 5.
- D. The program does not compile.

5. A variable defined inside a method is referred to as _____.

- A. a local variable
- B. a block variable
- C. a global variable
- D. a method variable

6. What will be displayed by the following code? _____.

```
class Test {  
    public static void main(String[] args) {  
        int[ ] list1 = {3, 2, 1};  
        int[ ] list2 = {1, 2, 3};  
        list2 = list1;  
        list1[0] = 0; list1[1] = 1; list2[2] = 2;  
        for (int i = list2.length - 1; i >= 0; i--)  
            System.out.print(list2[i] + " ");  
    }  
}
```

- A. 2 1 0
- B. 1 2 3
- C. 0 1 2
- D. 3 2 1

7. What is the output of running class C? _____.

```
class A {  
    public A( ) {  
        System.out.println("The constructor of A is invoked");  
    }  
}  
  
class B extends A {  
    public B( ) {  
        super( );  
        System.out.println("The constructor of B is invoked");  
    }  
}
```

```

    }
}
public class C {
    public static void main(String[] args) {
        B b = new B( );
    }
}

```

- A. none
- B. "The constructor of B is invoked"
- C. "The constructor of A is invoked" followed by "The constructor of B is invoked"
- D. "The constructor of A is invoked"

8. Analyze the following program, which description is correct _____.

```

class Test {
    public static void main(String[] args) {
        try {
            String s = "5.6";
            Integer.parseInt(s);
            int i = 0;
            int y = 2 / i;
            System.out.println("Welcome to Java");
        }
        catch (Exception ex) {
            System.out.println("Exception Done");
        }
    }
}

```

- A. An exception is raised due to Integer.parseInt(s);
- B. An exception is raised due to 2 / i;
- C. The program output "Welcome to Java";
- D. The program compiles and runs without exceptions.

9. What modifier should you use on a member variable so that it can only be referenced inside its defining class. _____

- A. public
- B. private
- C. protected
- D. Use the default modifier.

10. Which description about layout is **incorrect** _____.

- A. FlowPane arranges the components in the pane horizontally from left to right.
- B. GridPane arranges the components in the pane in a grid formation.
- C. BorderPane can place components in five regions: top, bottom, left ,right and center.
- D. HBox lays out components in a vertical columns.

Part II: Write Programs.(total 70 pts)

1. (10 pts) Write a method that will display **odd** numbers from **1** to **n**, 7 numbers per line. The numbers are separated by one space. The method header is as follows:

public static void displayOddNumber(int n)

2. (15 pts) Write a loop that computes the sum of below expression and displays the result reserving two decimals:

$$\frac{100}{1} + \frac{99}{2} + \frac{98}{3} + \frac{97}{4} \dots + \frac{3}{98} + \frac{2}{99} + \frac{1}{100}$$

3. (20 pts) Define an abstract **Geometry** class, which contains:

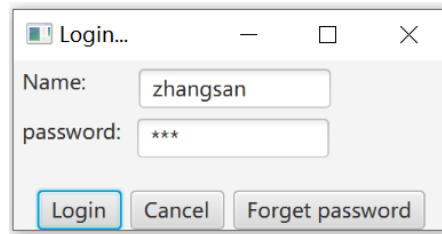
- Two member variables named **center_x** and **center_y** that specify the center of the geometry object on the 2D coordinate system.
- A constructor.
- An Abstract method **contain(int x, int y)** that returns whether the geometry contains the point (x, y)

Define a subclass **Rectang2D** that extends the **Geometry** class and contains:

- Two private member variables named **length** and **width**.
- A constructor.
- The **get-** and **set-** methods to access the private member variables.
- The overridden **contain(int x, int y)** method.

Write a test program, and create a **Rectang2D** object and check whether the **point (2,4)** is inside the rectangle.

4. (25 pts) Write a program to permit user to login in a system, as shown in below figure.



Requirements:

- Use suitable Java FX GUI components.
- Use suitable layout to organize components.
- At least contains one event handling process.